

Exploring the Relationship between Stellar Mass, Star Formation Rate, and Gas Outflow Velocity in a Large Sample of Galaxies

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Abstract

Galactic outflows are thought to be an important mechanism for reducing the efficiency of star formation in galaxies. We present new results on the relationship between stellar mass, star-formation rate, and gas outflow velocity in a large sample of galaxies. In particular, we use measurements of hydrogen emission lines to estimate the star-formation rate and dust content for a sample of 300,000 galaxies. These measurements will inform subsequent work with spectral stacking to explore how outflow properties depend on galaxy properties.

CHAPTER 1

Introduction

1.1. Background on galaxies and star formation

Our ever expanding universe is composed of hundreds of billions of galaxies. These galaxies, including our own Milky Way galaxy, contain dark matter, stars, gas clouds, black holes, and dust. When it comes to stars, many of us have heard of galaxies, which consist of stars, dust and matter, and when it comes to stars, they are formed when clouds of cool gas begin to collapse under a gravitational attraction, and when the core is dense enough, hydrogen begins to fuse into helium in the core. We can conceptualize them as luminous spheroids of plasma held together by its own gravity on a macroscopic scale.

The process of stars being formed, or star formation rate, is surprisingly very inefficient due to outflows. We are focusing on gas outflow velocity. It is common thought that it is the reason why SFR is so inefficient. In her senior thesis, my colleague Natasha Jones had to address why about five percent of the material used to form stars is actually utilized and states, “We have observed that many large galaxies either form fewer stars than we might expect or suddenly stop forming stars altogether. ” (Jones 2022).

1.2. Theoretical and Observational Work on Galaxy Out flows

More recent work has shown that stellar feedback can explain inefficient star formation across a wide range in galaxy mass (Hopkins et al., 2014). A number of theoretical models have been suggested to reduce the efficiency of star formation in lower-mass galaxies (primarily

negative feedback from supernova explosions) and high-mass galaxies (including feedback from accreting super-massive black holes, Somerville et al., 2008).

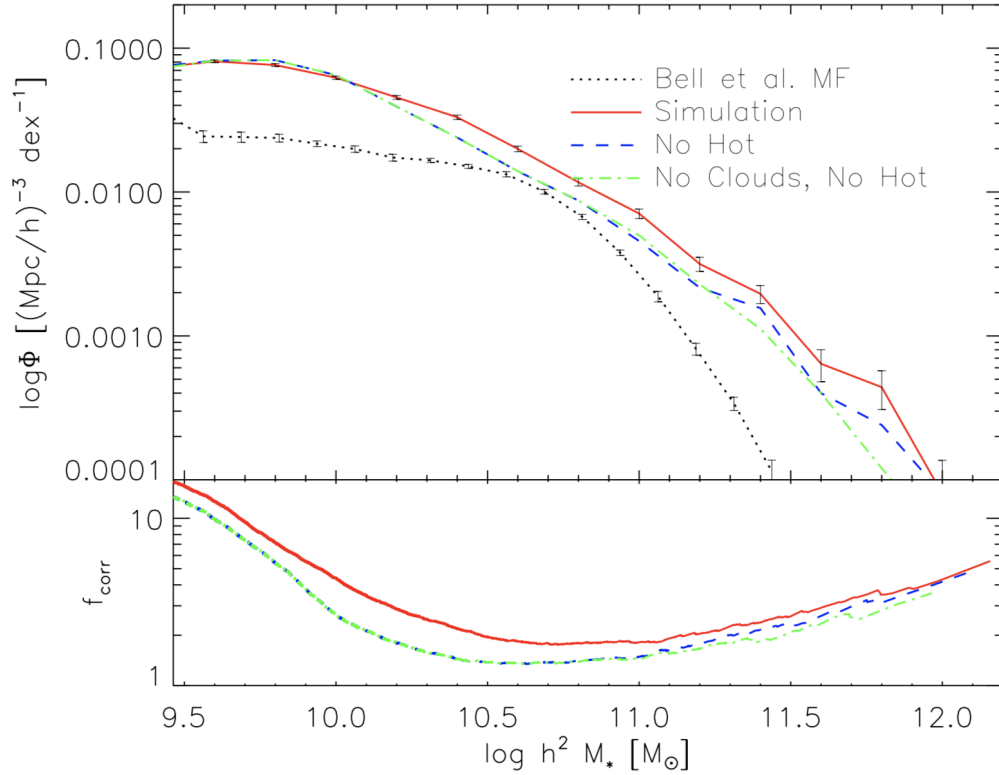


FIGURE 1.1. This figure from Kereš et al. (2009) shows how simulations without strong feedback overpredict the stellar mass of galaxies at both the low mass and high mass end of the stellar mass function. These are simulations of galaxy formation that follow how gas cools, collapses, and forms stars within dark matter halos, and they vastly overpredict the number of stars that form in galaxies. The implication is that feedback in the form of outflows is necessary to explain why the stellar mass is only about 5 percent of the baryonic mass in the universe.

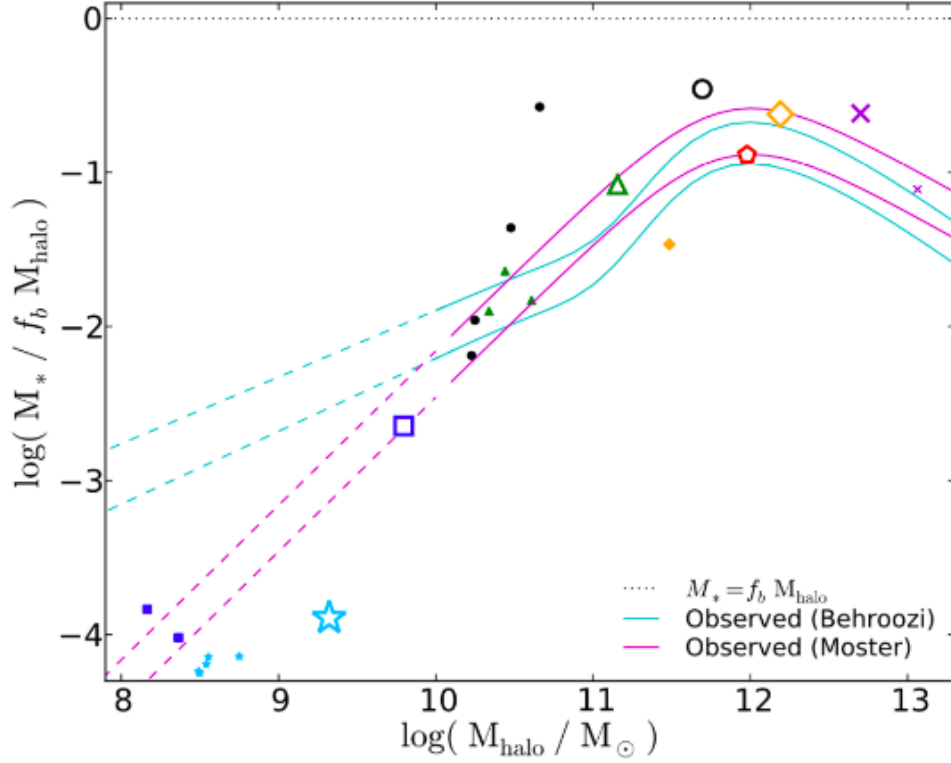


FIGURE 1.2. This figure from Hopkins et al. (2014) shows the result of high-resolution numerical simulations of galaxy formation in which stellar feedback reduces the efficiency of star formation. The vertical axis shows the fraction of the available supply of baryonic matter that forms stars in a galaxy on a logarithmic scale (i.e., -1 is 10 percent, -2 is 1 percent). The simulations match the relationship between stellar mass and dark matter halo mass found in previous studies.

There have been numerous observational studies conducted on gas outflows. Of the variety of papers that could be considered, some are the following: Weiner et al. (2009), which describes stacked spectra showing outflows at $z \sim 1.4$; Martin et al. (2012), displaying individual spectra of hundreds of galaxies; Rubin et al. (2014), showing individual spectra of hundreds of galaxies; and Zhu et al. (2015), which showcases the first stacked spectrum in eBOSS.

1.3. Context for this thesis

At some point in our lives, we have heard of phrases referring to something being “bright like a star,” or in the context of this thesis, the luminosity, or light of a star, is the

thermonuclear fusion of hydrogen into helium in their core. (Valdez 2019) Consequently, if a galaxy is composed of hundreds of millions of stars, which have an aforementioned luminosity characteristic, we can also describe galaxies as having a luminosity, and this will simply be one of the tools that will assist us in this study. We will be presenting the relationship of characteristics that can describe galaxies, such as displacement velocity, stellar mass, star formation rate, source star formation rate, and max velocity to name a few. There will be a focus on a large sample of three million galaxies, and since only ten percent of them are driving these outflows, our sample will consist of three hundred thousand galaxies from the eBOSS survey (Extended Baryon Oscillation Spectroscopic Survey), an “observation of galaxies and quasars, in a range of distances (red-shifts) currently left completely unexplored by other three-dimensional maps of large-scale structure in the Universe.” This study will focus on the relationship between stellar mass, star formation rate, and gas outflow velocity. Its value will act as an introduction to understanding the characteristics of galaxies and how complex their relationships can be to discover. This thesis will consist of data extracted from the eBOSS and Python that is used to make interpretations of the relationships between the aforementioned characteristics.

CHAPTER 2

Background

2.1. Atomic physics and the H α emission line

Before talking about these characteristics, we need to start off somewhere comfortable. The term Hydrogen-Alpha (H α) may be familiar to some readers, but here is a description for some who do not know.

There are multiple energy levels, but our focus is on the second energy level. When an electron of a hydrogen atom transitions to the second energy level, this is called the Balmer series, it produces a photon with a specific wavelength corresponding to the energy difference between the energy levels. The transition of focus is from the third energy level. As E3, denoting a hydrogen atom's third energy level, transitions to E2, there is an electromagnetic emission with a frequency of about 656.28 nm, or 6562.8 Å. The series of possible transitions down to n=2 (from n=3, 4, 5, etc.) is the aforementioned Balmer series.

An emission spectrum line display will feature a series of spectral lines that are used by, in this case, astronomers to detect atoms, molecules. “H-Alpha,” or H α , is a specific deep-red visible spectral line in the Balmer series. This series is important for our work in large part because it consists of wavelengths that are visible to the human eye and observable by ground-based optical spectrographs.

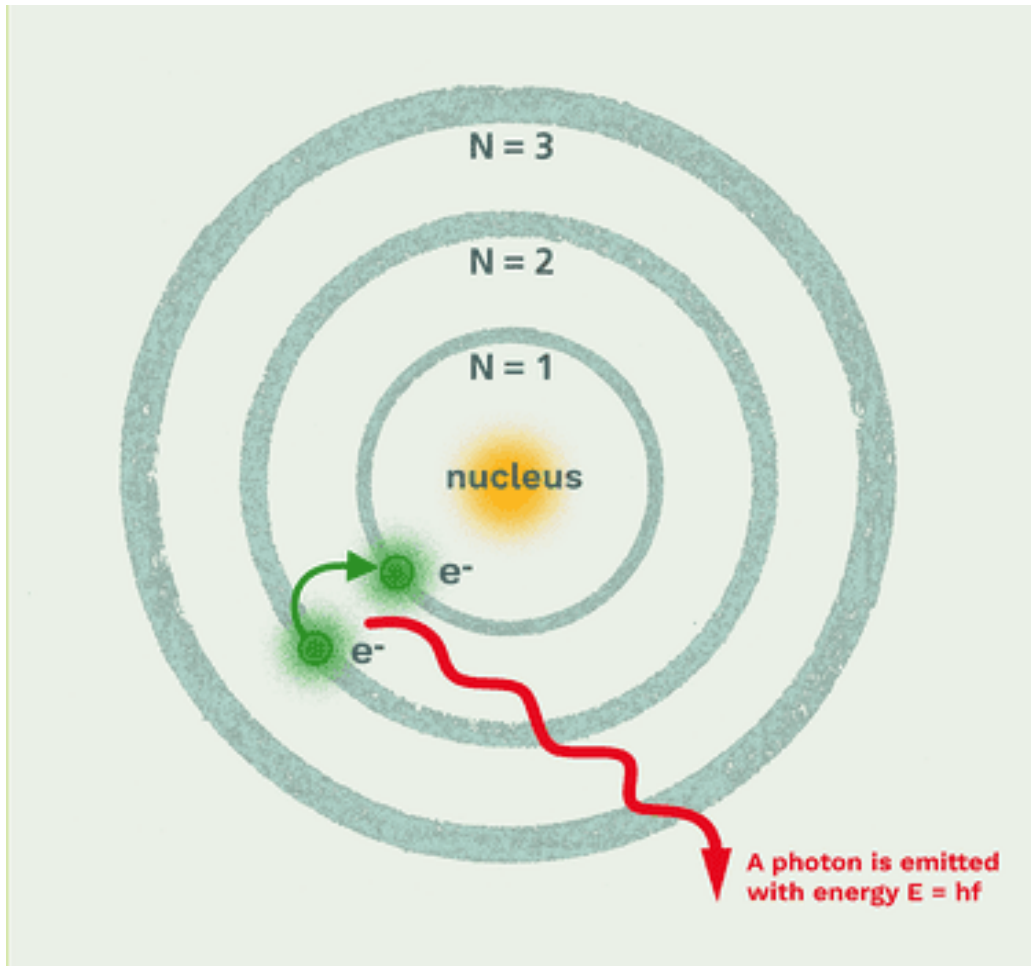


FIGURE 2.1. Figure showing how an electron produces an emission-line photon.

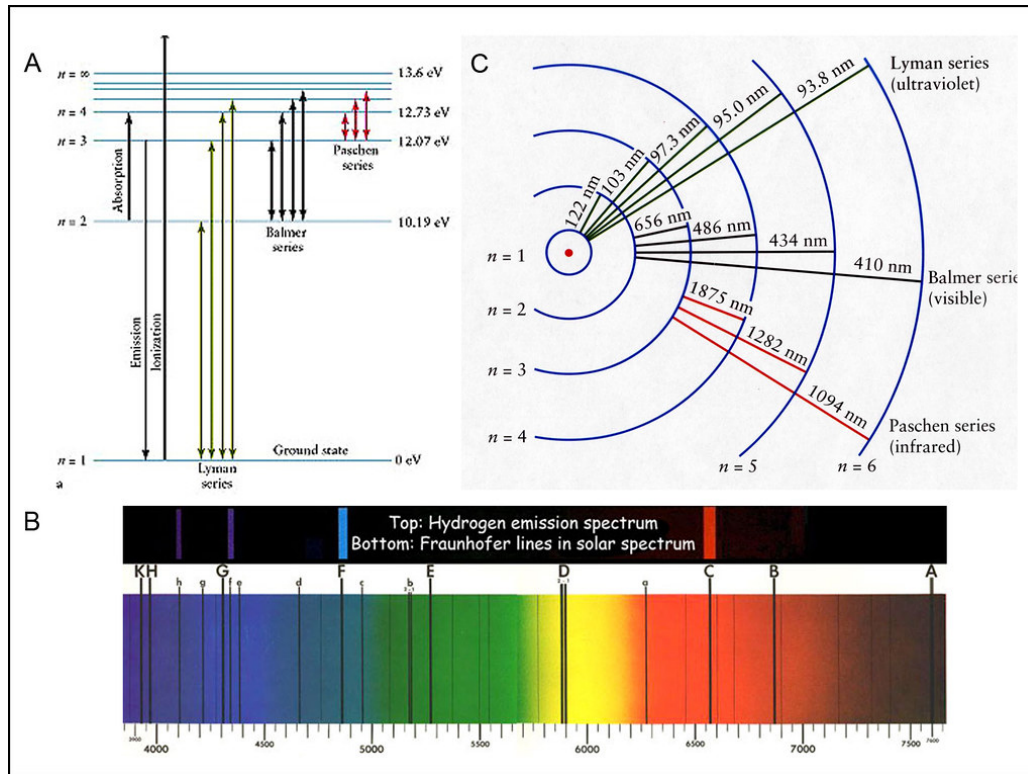


FIGURE 2.2. Figure showing the various hydrogen transitions, including the Balmer series.

2.2. How we use H α to estimate star-formation rate

The study of H α ties with star formation rate with equations that derived in Kennicutt & Evans (2012). The following equation is:

$$(1) \quad \log_{10}(SFR) = \log_{10}(L_{H\alpha}) - 41.27$$

We can use H α to help approximate the number of ionizing photons produced per second in the galaxy because it acts as a balance between photo-ionization and recombination in the ionized hydrogen regions (called H-II regions) near where stars have recently formed.

2.3. Using emission-line ratios to estimate dust attenuation

We can also use the ratio of emission lines to determine the necessary dust attenuation, dust correction, due to galactic dust impacting shorter wavelength light greater than higher wavelength light. There is considerable significance to this because our data loses accuracy without it. When we are making measurements between our planet and stars, there are tons of galactic dust that absorbs light, meaning that whatever we observe is fainter than it intrinsically is. This means our intrinsic measurements should be lower than the observed measurements. H α and H β correspond to red and blue light, respectively. The ratio of flux between H α and H β is 2.86:1, which is established by atomic physics. This means H α is about three times stronger than H β , and this ratio acts as a fundamental intrinsic feature of them. Consequently, dust attenuation affects blue light measurements more than red.

2.4. Why emission-line luminosity is connected to star-formation rate

The mass of a galaxy can be characterized by stellar mass. Hydrogen atoms can absorb light at specific frequencies. In space, when a beam of light is shown through a gas, its many

colors, sometimes resembling a rainbow, will get absorbed from the beam. This beam of light will consist of what astronomers call absorption lines. Each atom and molecule will have distinct absorption fingerprints, or absorption lines, thus allowing astronomers to tease out the chemical makeup of distant stars and galaxies. Hydrogen atoms can also produce emission lines. These are observed spectral lines that are due to the electron making transitions between two energy levels in an atom. Spectral emission occurs when an electron transitions from a higher energy state to a lower energy state. Through this transition of energy levels, a photon is emitted by the atom. Using emission line luminosity, the star formation rate can be found using the following steps:

- (1) We have a finite total number of recently formed stars.
- (2) A small fraction of those stars are O-type stars that are able to produce loads of photons with wavelengths less than 912 Angstroms and energies greater than 13.6 eV.
- (3) These ionizing photons propagate outwards and ionize all of the hydrogen in some region, sometimes called a Stromgren sphere. After each ionization, there is a subsequent recombination and the electron cascades down to the ground state with some probability of producing an H α photon.
- (4) We observe the energy output per second of H α to infer how many ionizing photons are being produced, and we then can infer the quantity of O-type stars and total number of recently formed stars.

We are essentially measuring the number of O stars when we measure H-alpha luminosity and we require a stellar population model to determine how many total stars have formed (not just the most massive O stars). Side tangent: Stars are categorized as a type in a surface temperature sequence called OBAFGKM. This does not stand for “O Be A Fine Girl, Kiss Me...” it represents the following:

Spectral Type	Color	Temperature (K) [*]	Spectral Features
O		28,000-50,000	Ionized helium, especially helium
B		10,000-28,000	Helium, some hydrogen
A		7,500-10,000	Strong hydrogen, some ionized metals ^{**}
F		6,000-7,500	Hydrogen and ionized metals such as calcium and iron
G		5,000-6,000	Both metals and ionized metals, especially ionized calcium
K		3,500-5,000	Metals
M		2,500-3,500	Strong titanium oxide and some calcium

^{*} To convert approximately to Fahrenheit, multiply by 9/5.
^{**} Astronomers regard elements heavier than helium as metals.

FIGURE 2.3. Image credit: Sky & Telescope. This shows the spectral classes or spectral types of stars, with the most massive stars being O stars. These hottest stars produce significant ultraviolet radiation that can ionize hydrogen.

2.5. Redshift, distance, and lookback time in an expanding universe

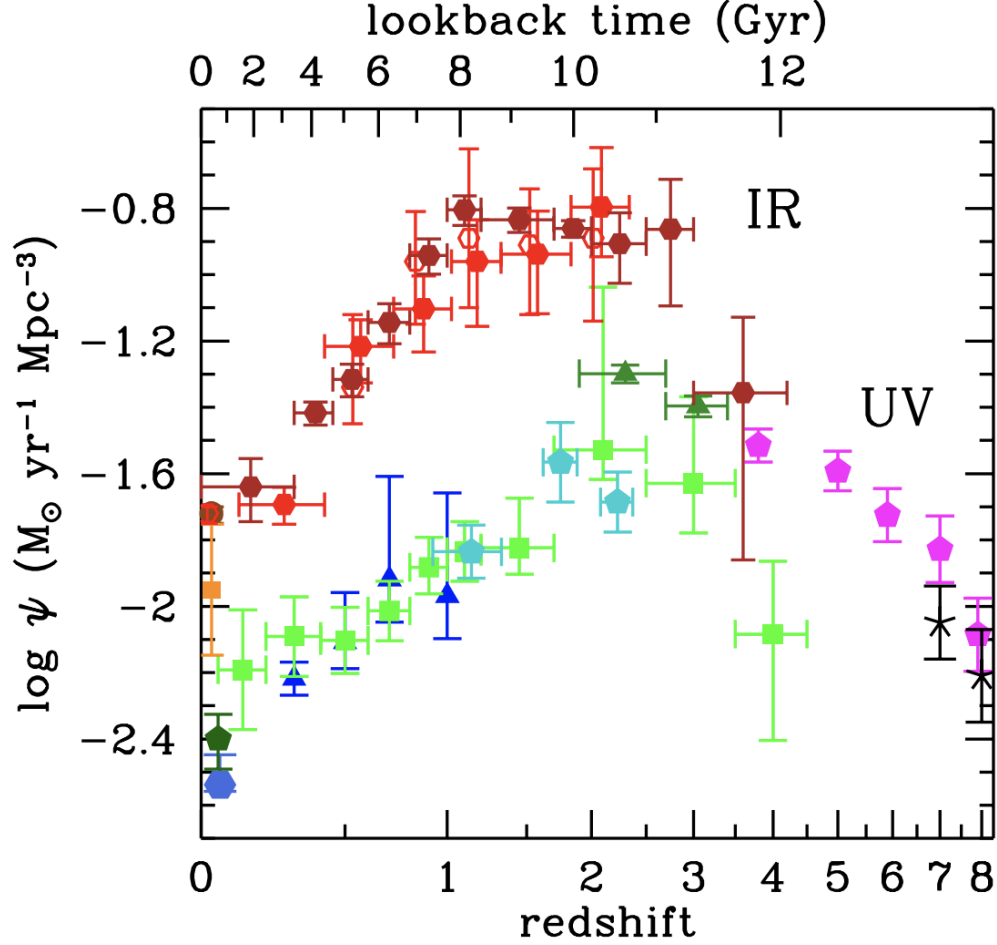


FIGURE 2.4. This figure from Madau & Dickinson (2014) shows a compilation of observational results that have used observations at infrared and ultraviolet wavelengths to estimate the volume density of star formation in the universe. This star formation density rises for larger lookback times and larger redshifts out to approximately $t=10$ Gyr and $z=2$. Our sample spans a range between $z=0.4$ (lookback time, $t=4$ Gyr) and $z=1.2$ (lookback time, $t=8$ Gyr).

We are describing how long ago the light from a source at a specified red-shift, z , was emitted using the following equations:

$$(2) \quad z = \frac{\lambda_{obs}}{\lambda} = \frac{\lambda_{emit}}{\lambda}$$

$$(3) \quad t(z) = \int_0^z 1/H(z) dz$$

where $t(z)$ is the look-back time and $H(z)$ is the Hubble parameter.

There is H_0 , which is the present value of the Hubble parameter, which sets the scale of our universe at present time using scale factor a ,

$$(4) \quad H_0 = H(z = 0) = 70 \frac{km/s}{Mpc}$$

Where $a = 1/(1+z)$ and H can be described as $\dot{a}/a$. If a scientist wants to describe the scale factor of today, a would be equal to 1. The Hubble constant is not actually a constant, but is regarded as measuring the expansion rate today.

$$(5) \quad H(z) = H_0 * \sqrt{[(\Omega_A + \Omega_m(1+z)^3 + \Omega_r(1+z)^4)]}$$

where Ω_A represents the present amount of dark energy in the universe, $\Omega_m(1+z)^3$ represents the present amount of matter in the universe, and $\Omega_r(1+z)^4$ represents the present amount of how much radiation. These are all ratios, so they are unit-less. The value of Ω_x can be described as

$$(6) \quad \Omega_x = \frac{\rho_x}{\rho_c}$$

where ρ_x describes the density of a particular characteristic like dark matter, matter, or radiation; and ρ_c represents critical density that represents a flat universe. As I am writing this, the amount of dark energy, matter, and radiation in the universe is 0.7, 0.3, and 10^{-4} .

CHAPTER 3

Data and Sample

3.1. eBoss Data

Our data comes from the Extended Baryon Oscillation Spectroscopic Survey (eBOSS). The eBOSS survey is a part of the Sloan Digital Sky Survey (SDSS) meant to observe galaxies and quasars across red-shifts ranging from $z=0.5$ to 3. All spectra are observed using the BOSS spectrographs. The spectrographs both have a red camera and a blue camera that allows for the total wavelength coverage of the detector to be 360–1000 nm. The resolution is $R \sim 2000$. In total the survey has observations of more than two million galaxies and quasars.

3.2. Sample

For our analysis we are only interested in galaxies that are still forming stars and are of a red-shift between $z < 0.4$ and $z > 1.2$. Therefore, we are only interested in a subset of the galaxies within the eBOSS survey. Our data sample includes 300,000 of galaxies which is approximately 10% of all targets observed in the eBOSS survey. With those 300,000 sources we sort them by physical properties such as their star-formation rate (SFR) using their observed hydrogen emission lines. In recent years it has become increasingly clear that “feedback” in the form of outflowing gas and dust is a crucial aspect of galaxy evolution, explaining why galaxies form so few stars (e.g., Somerville & Davé, 2015). The spectral features of galaxies are one of a few different probes used to understand these outflows. The figure below shows an optical spectrum created by stacking the spectra of 12,000 individual galaxies.

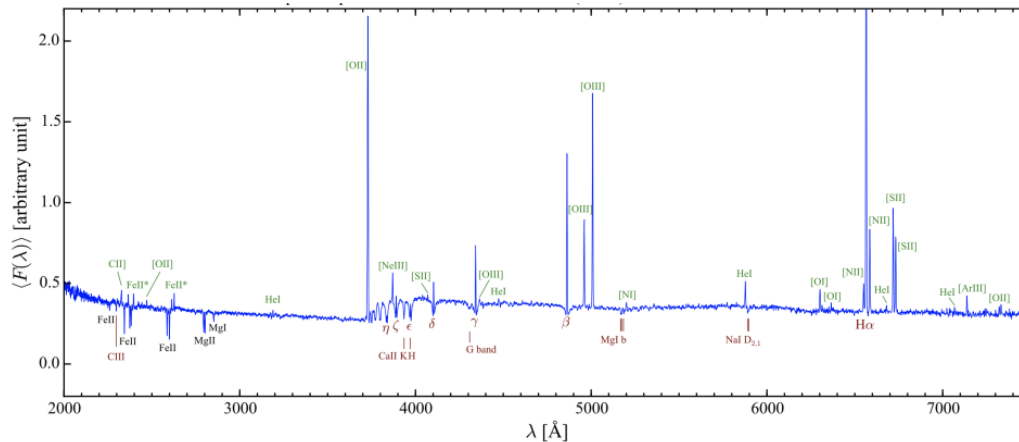


FIGURE 3.1. Figure from Zhu et al. (2015) showing a stacked spectrum from the first 12,000 galaxies observed in the eBOSS survey. The ultimate goal of our larger project is to create stacked spectra like this one in bins of physical quantities such as star-formation rate.

The data predating this one stems from the FITS format and features a HIZEA FIT table that has 50 compact starburst galaxies with fast outflows in previous years, including the aforementioned characteristics, galaxy names, and red shifts, represented by the letter Z. Astronomers utilize this to measure how the universe is expanding, and thus to determine the distance between us and our universe's most distant, and therefore oldest, objects. The other is the eBOSS which was mentioned earlier. The HIZEA survey and eBOSS feature data on galaxies and differ in the red-shifts and sample sizes of their galaxies. HIZEA features extreme galaxies, compact starburst galaxies with fast outflows, and consists of about fifty galaxies. eBOSS features stacked spectra, can describe how outflow properties vary as a function of galaxy properties, and consists of about three millions galaxies, which includes the approximate three hundred thousand galaxies. The eBOSS data is compiled from the eBOSS spectrograph that measures the optical spectra of 1,000 different objects simultaneously. By identifying spectral features (e.g., emission lines, absorption lines) one can use the spectra to measure the cosmic redshift of each galaxy. Additionally one can also use those features to estimate things like star-formation rate, dust content, and outflow velocity. If we observe

galaxies with a red-shift greater than 0.5, a wavelength that exceeds the range of the eBOSS spectrograph, 5650-10K Angstroms, is being utilized meaning that galaxies that exceed this red-shift cannot be detected. There are two different types of wavelengths: rest and observed wavelengths. “The rest wavelength of a source of electromagnetic radiation is the wavelength as observed if at rest in relation to the radiation source. If source and observer are in relative motion, the wavelength observed is termed the observed wavelength.” On Earth, there is no difference between observed and rest wavelengths. The focus of this thesis will be on data from the eBOSS.

3.3. Additional context for why studying out flows is important

The study of outflows is vital for understanding the universe due to galaxies being inefficient at converting gas into stars. We believe that one of the reasons relates to these outflows that are injecting energy and momentum into the gas supply. These studies include observations in gas velocity of absorption lines, gas velocity from emission lines, gas column density from absorption lines, and the geometry of the outflow. This also has utility in estimating mass outflow rate, gas temperature, gas density, and a geometric, opening angle of a galaxy. Galactic outflows were first found in nearby starburst galaxies (Axon & Taylor, 1978; Rieke et al., 1980), but we now know that they are prevalent in galaxies at all red-shifts (e.g., Shapley et al., 2003; Weiner et al., 2009; Chen et al., 2010; Martin et al., 2012).

Galaxy formation is surprisingly inefficient, in that galaxies have much less stellar mass than would be indicated by the cosmic baryon fraction (e.g., Moster et al., 2013), which is defined as the “amount of hydrogen in the Universe today, either in stars and galaxies, or hot gas between galaxies. The current measurements indicate that the baryon fraction is about 5 percent (0.05) the value of closure for the Universe (the critical density).” In recent years

it has become increasingly clear that “feedback” in the form of out-flowing gas and dust is a crucial aspect of galaxy evolution, explaining why galaxies form so few stars (e.g., Somerville & Davé, 2015). Unfortunately, the physical mechanisms for feedback are poorly constructed due to a large quantity of processes including supernova explosions, radioactive feedback (radiation pressure, photo-ionization, photoelectric heating), and stellar winds. An additional challenge has been the complex multi-phase structure of galactic winds and the diffuse nature of the gas they contain (Veilleux et al., 2020). Another has been a fragmented observational approach to studying galactic winds through down-the-barrel spectroscopy, which provides unambiguous kinematic information, but shares no direct information on the radial extent of the outflow because we do not know how it is distributed along the line of sight. This type of spectroscopy, down-the-barrel spectroscopy, means examining absorption lines associated with gas in between a viewer and the source, such that she can interpret blue-shifted absorption lines as tracing out-flowing gas that is moving towards her (with the shift in wavelength proportional the outflow speed via the Doppler effect).

3.4. Why study out ow properties as a function of galaxy properties?

There is value in studying how outflow properties vary as a function of galaxy properties. We know that galaxy formation is inefficient at all mass scales, but galaxies at the low-mass end and the high-mass end of the stellar mass function have the lowest ratios of stellar mass to baryonic mass. If outflows are responsible for explaining this behavior, then we may expect to see evidence for stronger outflows in galaxies that are the most inefficient at forming stars. In addition, different physical models for outflows implemented in numerical simulations of galaxy formation have different predictions for how the mass outflow rate and outflow speed should vary with stellar mass or star formation rate. For example, some models predict that

outflow velocity should scale with stellar mass (e.g., Oppenheimer & Davé, 2006) and others predict that outflow velocity should be highest in galaxies with the highest surface density of star formation (e.g., Heckman et al., 2015).

We are planning to measure physical properties of these galaxies, such as stellar mass and star-formation rate, using the emission-lines. The SFR will be measured using H α luminosity. It will connect to subsequent work that also measures stellar mass and performs spectral analysis in groups of SFR and stellar mass to measure outflow velocity based on stacked spectra. The specific focus of this thesis is establishing the framework for the subsequent work.

CHAPTER 4

Results

In order to focus on the galaxies with significant detection of both H α and H β we filter them using their respective signal-to-noise ratios to evaluate the quality of the data. The signal-to-noise ratio of H α and H β both exceeded three. For the H α sample, the red-shifts ranged from $0.4 < z < 0.58$, and for the H β sample, the red-shifts ranged from $0.58 < z < 1.2$.

The simple atomic structure of hydrogen has allowed observers to determine the expected rates of transition and the emission line ratios as a function of the temperature and electron density of the interstellar medium. Usually, when determining intrinsic line ratios, it is assumed that the nebula is optically thick to photons in the Lyman series but optically thin to Lyman and photons of the higher hydrogen series. Under this assumption, the expected intrinsic ratio of H β to H α is

$$(7) \quad \frac{H\beta}{H\alpha} = 2.86$$

from Osterbrock (1989). Dust interfering with the ionization will impact our measured emissions. The effect of dust is to diminish the observed emission of the short-wavelengths, H β , more than the long wavelengths, H α . The attenuated observed line ratios will inherently be lower than the intrinsic measurements when dust is present. We follow the deviation of dust correction as presented in Momcheva et al. (2013). The stronger attenuation of is demonstrated in a color excess as:

$$(8) \quad A(H\gamma) - A(H\beta) = E(H\gamma - H\beta) = -2.5 * \log_{10} \left[\frac{(H\beta/H\gamma)_{int}}{(H\beta/H\gamma)_{obs}} \right]$$

In context of this equation, $(H\beta/H\gamma)$ acts as the observed ratio. The color excess is related to the broadband color excess $E(B-V)$ via the attenuation curve:

$$(9) \quad A(\lambda) = k(\lambda)E(B - V)$$

where $k(\lambda)$ is the corresponding attenuation curve at the wavelength λ . Given we are interested in computing $E(H\gamma - H\beta) = E(B - V)[k(H\gamma) - k(H\beta)]$. It consequently follows:

$$(10) \quad E(B - V) = \frac{A(\lambda)}{k(\lambda)} = \frac{2.5}{k(\lambda_H) - k(\lambda_{H\beta})} * \log_{10} \frac{(H\beta/H\gamma)_{int}}{(H\beta/H\gamma)_{obs}}$$

Given $k(H\gamma) = 3.33$ and $k(H\beta) = 4.60$, we have an equation that can describe the desired attenuation:

$$(11) \quad A(H\gamma) = \frac{2.5}{3.33 - 4.60} \log_{10} \frac{2.86}{(H\beta/H\gamma)_{obs}} * 3.33$$

Using this attenuation, we can compute a value for $H\gamma$ correction using

$$(12) \quad H\gamma_{corr} = 10^{A(H\gamma)/2.5}$$

Given we have already measured the $H\gamma$ immensities, the last task is computing $H\beta$ immensities and use the ratio to evaluate the presence of dust.

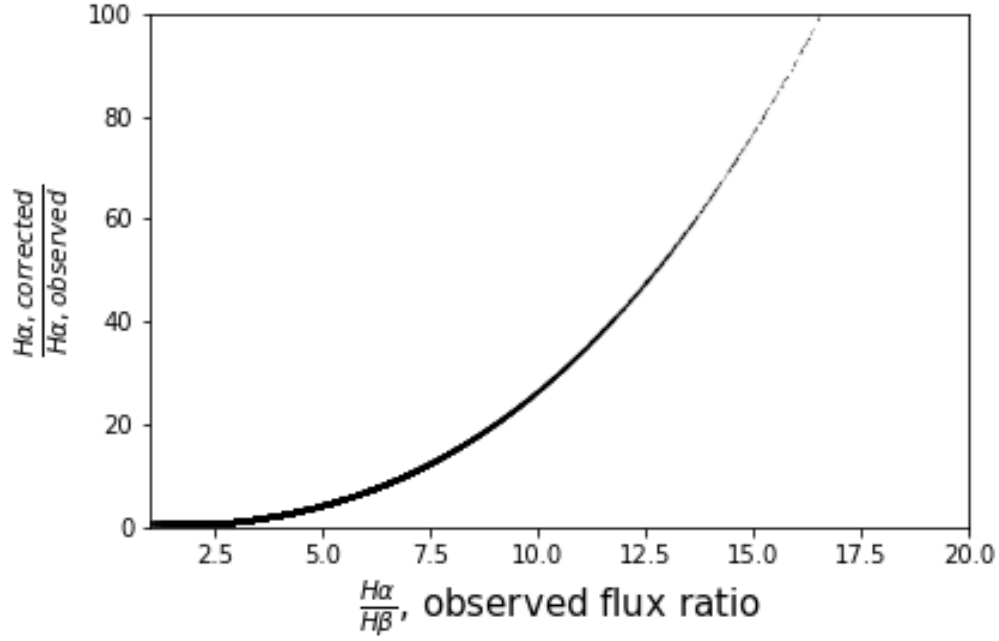


FIGURE 4.1. The H-alpha/H-beta ratio is directly related to dust, which absorbs shorter wavelength light more than longer wavelength light. In this plot, large amounts of dust is absorbing light at shorter wavelengths.

4.1. Calculation of H α dust correction and intrinsic H α luminosity

$$(13) \quad F_i = F_o \times 10^{0.4k(\lambda)E(B-V)}$$

The visualization in Figure 4.1 describes the H α dust correction as a function of the H α /H β ratio. This shows how the correction factor is larger for galaxies with larger H α /H β ratios. This is because the larger ratios are caused by dust reddening. From this, we know that we need to apply correction factors as large as 100 $\times$ to obtain the intrinsic H α flux. We also show in Figure 4.2 how the corrected H α flux compares to the observed flux to give a sense for the impact of this correction on our large sample of galaxies. For reference, we include a line that corresponds to the corrected H α being about ten times larger than the observed flux. Finally, we show in Figure 4.3 how the galaxies with the largest H α luminosities (and therefore the

largest SFR values) also tend to be the galaxies that are dustier (i.e., they tend to have larger H α /H β ratios).

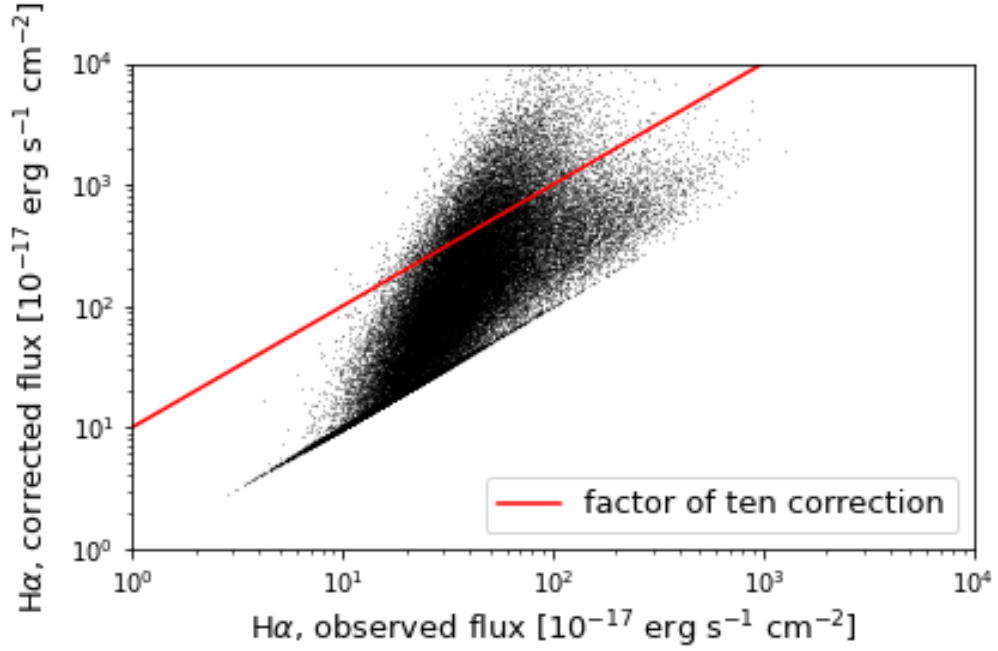


FIGURE 4.2. Displays how the H-alpha flux for each galaxy was corrected for dust attenuation.

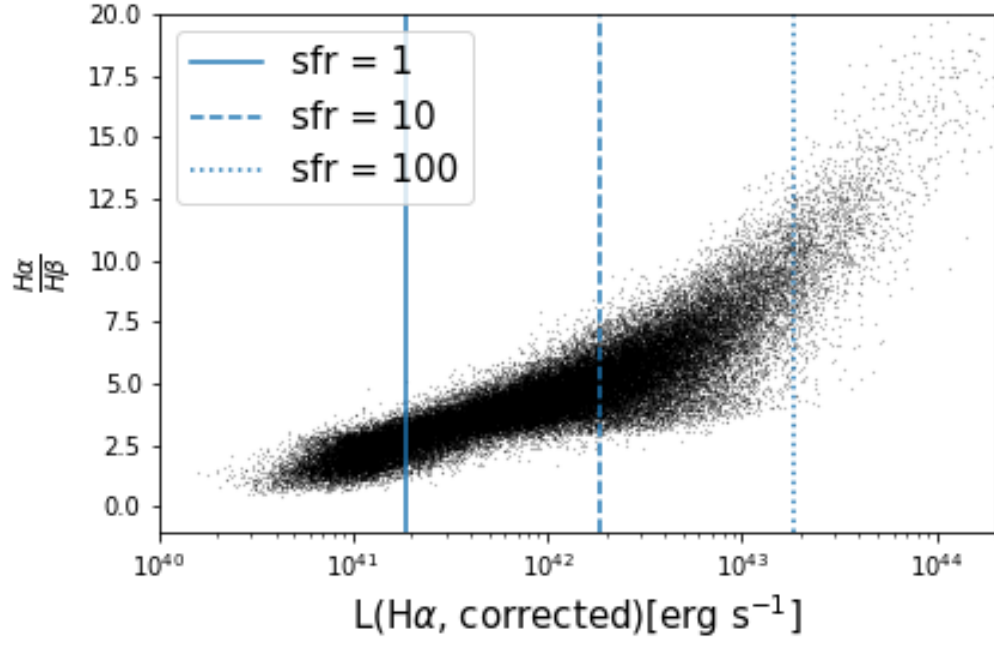


FIGURE 4.3. The luminous sources of H-alpha emission, the sources with the highest star formation rate are also dustier, meaning there is more dust attenuation in those galaxies. This means the galaxies that are forming the most stars are the dustiest. The galaxies with the highest SFR are the dustiest.

4.2. Calculation of $H\beta$ dust correction and luminosity for galaxies at $z > 0.58$

Because the observed $H\beta$ lines moves outside the range of the spectrograph for galaxies with red-shifts $z > 0.58$, we need to find a way to use the other Balmer lines to estimate SFR. In particular, we are able to determine $H\beta$ intensities by using the $H\beta/H\gamma$ ratio to calculate a dust correction. We show the dust correction factor for this method in Figure 4.4 and we show the relationship between corrected and observed flux in Figure 4.5. Similar to the results for $H\alpha$, we find that the galaxies with the largest $H\beta$ intensities (and therefore star-formation rates) are also the dustiest (see Figure 4.6).

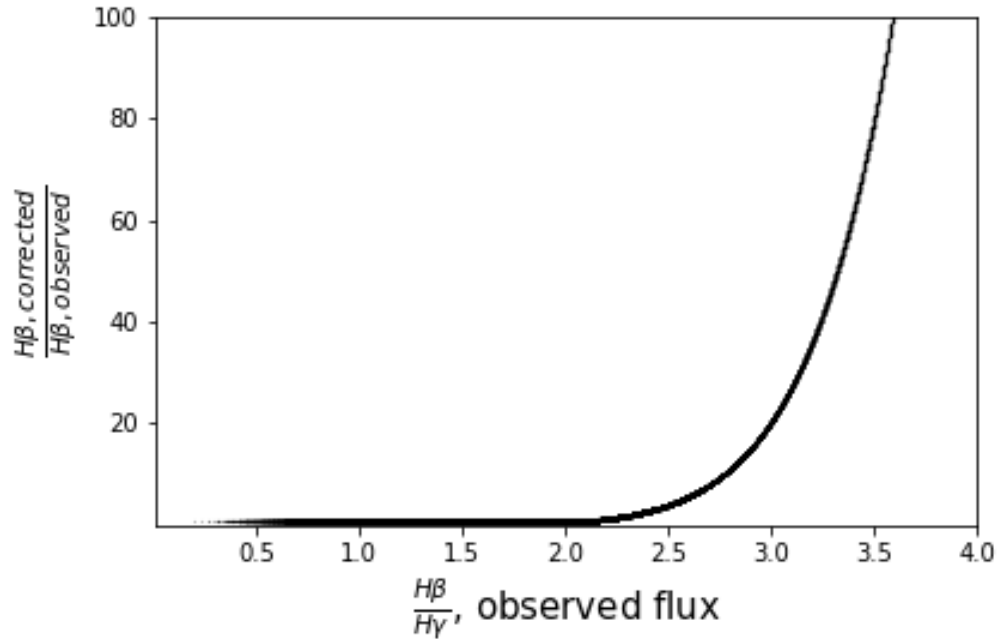


FIGURE 4.4. The $H\beta$ dust correction factor as a function of the observed $H\beta/H\gamma$ ratio.

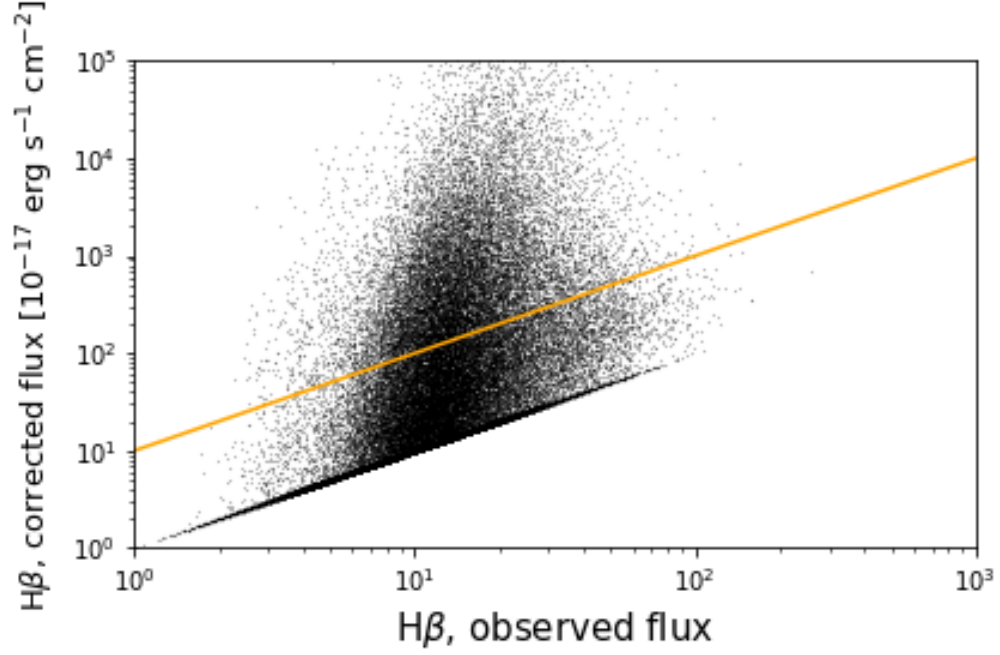


FIGURE 4.5. The relationship between the dust-corrected $H\beta$ flux and the observed $H\beta$ flux.

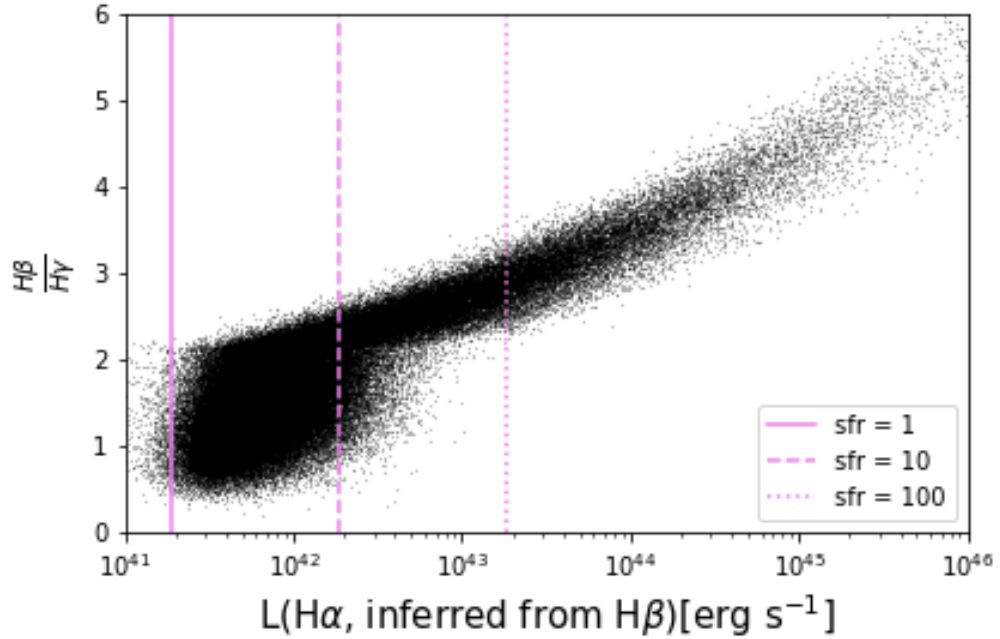


FIGURE 4.6. The relationship between the amount of dust reddening, traced by the $H\beta/H\gamma$ ratio, as a function of the $H\beta$ luminosity. Here we multiply the dust corrected $H\beta$ luminosity by a factor of 2.86 as a proxy dust corrected H luminosity.

CHAPTER 5

Discussion and Conclusion

In the starting stages, we researched how we can use H to estimate star-formation rate and using emission-line ratios to estimate the proper dust attenuation (see Equation 1). We decided the best online database to gather information on galaxies given the focus of our red-shifts, sample size, and study on outflows would be the eBOSS survey. When we began this thesis, we knew that galaxy formation is inefficient at all mass scales, but especially in galaxies at the low-mass end and the high-mass end of the stellar mass function. We supposed that if outflows were responsible for explaining this behavior, then we may expect to see evidence for stronger outflows in galaxies that are the most inefficient at forming stars.

The primary result of this work is the calculation of H star-formation rates as demonstrated in Figure 4.3, where the straight, broken, and dotted lines shows the star formation rate of one, ten, and one hundred solar mass units per year. The next steps for the overall project (i.e., future work) involves stacking galaxies as function of star-formation rate in order to measure their outflow velocity. Ultimately, we want to quantify the relationship between these physical quantities (e.g., Chisholm et al., 2015).

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